MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, FLUORINATED ETHYLENE PROPYLENE, NON-CROSSLINKED

This specification is approved for use by all Department and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range: -67°C (-88°F) to +200°C (+392°F).

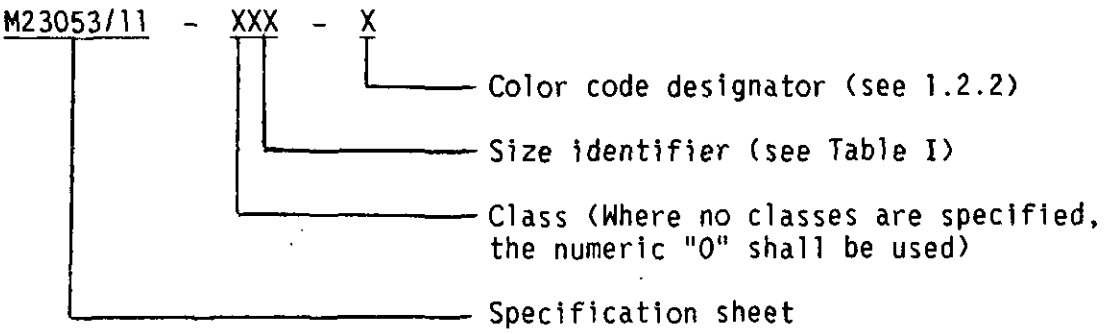
Color: The sleeving shall be furnished in clear (water white to transparent blue), the as manufactured color.

Class: The sleeving shall be furnished in the following classes, as specified (see 6.2.1):

Class 1 - Normal shrink ratio, Heavy Wall
Class 2 - High shrink ratio, Medium Wall

Longitudinal change: ± 15 percent

Military part number: The Military part number shall consist of this specification sheet and dash numbers as follows:



Example: Class 2, 0.750 inch (19.1 mm) as supplied ID sleeving shall be identified as M23053/11-207-C.

NOTE: Entire specification sheet revised. Class 1 and Class 2 are now further clarified as "Heavy Wall" and "Medium Wall", respectively.

AMSC N/A FSC 5970

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TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unresticted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/11-101-C	.031(.79)	.027(.68)	.008 ± .002(.20 ± .05)
M23053/11-102-C	.036(.91)	.032(.80)	.008 ± .002(.20 ± .05)
M23053/11-103-C	.045(1.1)	.039(1.0)	.008 ± .002(.20 ± .05)
M23053/11-104-C	.060(1.5)	.049(1.2)	.008 ± .002(.20 ± .05)
M23053/11-105-C	.075(1.9)	.061(1.6)	.009 ± .002(.23 ± .05)
M23053/11-106-C	.092(2.3)	.072(1.8)	.009 ± .002(.23 ± .05)
M23053/11-107-C	.115(2.9)	.089(2.3)	.009 ± .002(.23 ± .05)
M23053/11-108-C	.141(3.6)	.114(2.9)	.010 ± .003(.25 ± .08)
M23053/11-109-C	.158(4.0)	.124(3.2)	.010 ± .003(.25 ± .08)
M23053/11-110-C	.180(4.6)	.143(3.6)	.010 ± .003(.25 ± .08)
M23053/11-111-C	.197(5.0)	.158(4.0)	.011 ± .004(.28 ± .10)
M23053/11-112-C	.225(5.7)	.180(4.6)	.011 ± .004(.28 ± .10)
M23053/11-113-C	.248(6.4)	.198(5.0)	.011 ± .004(.28 ± .10)
M23053/11-114-C	.290(7.4)	.226(5.8)	.011 ± .004(.28 ± .10)
M23053/11-115-C	.310(7.9)	.249(6.4)	.011 ± .004(.28 ± .10)
M23053/11-116-C	.365(9.2)	.280(7.2)	.012 ± .004(.30 ± .10)
M23053/11-117-C	.400(10.2)	.311(7.9)	.012 ± .004(.30 ± .10)
M23053/11-118-C	.440(11.2)	.349(8.9)	.012 ± .004(.30 ± .10)
M23053/11-119-C	.500(12.7)	.383(9.7)	.015 ± .004(.38 ± .10)
M23053/11-120-C	.580(14.7)	.448(11.4)	.020 ± .004(.50 ± .10)
M23053/11-121-C	.666(16.9)	.510(13.0)	.020 ± .004(.50 ± .10)
M23053/11-122-C	.830(21.1)	.637(16.1)	.025 ± .004(.64 ± .10)
M23053/11-123-C	1.000(25.4)	.764(18.9)	.030 ± .004(.80 ± .10)
M23053/11-124-C	1.170(29.7)	.891(22.7)	.035 ± .004(.90 ± .10)
M23053/11-125-C	1.300(33.0)	1.000(25.4)	.020 ± .004(.50 ± .10)
M23053/11-126-C	1.330(33.8)	1.020(25.9)	.035 ± .004(.90 ± .10)
M23053/11-127-C	1.700(43.2)	1.300(33.4)	.020 ± .004(.50 ± .10)
M23053/11-128-C	2.100(53.3)	1.700(43.2)	.020 ± .004(.50 ± .10)
M23053/11-129-C	2.600(66.0)	2.100(53.5)	.020 ± .004(.50 ± .10)
M23053/11-130-C	3.100(78.7)	2.600(66.0)	.020 ± .004(.50 ± .10)
M23053/11-131-C	3.500(88.9)	3.100(78.7)	.020 ± .004(.50 ± .10)
M23053/11-132-C	4.300(109.2)	3.500(88.9)	.020 ± .004(.50 ± .10)
Class 2			
M23053/11-201-C	.093(2.4)	.056(1.4)	.008 ± .003(.20 ± .08)
M23053/11-202-C	.125(3.2)	.075(1.9)	.010 ± .003(.25 ± .08)
M23053/11-203-C	.188(4.8)	.115(2.9)	.010 ± .003(.25 ± .08)
M23053/11-204-C	.250(6.4)	.150(3.8)	.010 ± .003(.25 ± .08)
M23053/11-205-C	.375(9.5)	.225(5.7)	.012 ± .003(.30 ± .08)
M23053/11-206-C	.500(12.7)	.300(7.6)	.015 ± .004(.38 ± .10)
M23053/11-207-C	.750(19.1)	.450(11.3)	.020 ± .004(.50 ± .10)
M23053/11-208-C	1.000(25.4)	.600(15.0)	.025 ± .005(.64 ± .12)
M23053/11-209-C	1.500(38.0)	.900(22.5)	.030 ± .005(.80 ± .12)
M23053/11-210-C	2.000(51.0)	1.200(30.0)	.030 ± .005(.80 ± .12)

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1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

Unrestricted shrinkage: Test method 4.6.5, 175° ± 2°C (347° ± 4°F) for 10 minutes.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Restricted shrinkage	No cracks	4.6.6.1.1 175° ± 2°C (347° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Clarity stability	Pass	4.6.17 175° ± 1°C (347° ± 2°F), 24 hours
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength psi (MPa), minimum	2,000 (13.8)	4.6.14 ASTM D 638, 2 inches/minute
Ultimate elongation, percent, minimum	200	4.6.14 ASTM D 638, 2 inches/minute
Dielectric strength, volts/mil (Kv/mm), minimum	500(12.7)	4.6.2 ASTM D 2671
Volume resistivity, Ohm-cm, minimum	10^{17}	4.6.2 ASTM D 876
Water aborption, percent, maximum	0.01	4.6.2 ASTM D 570, 24 hrs at 23°C

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Low temperature flexibility	No cracks	4.6.7.1 -65° ± 1°C (-85° ± 2°F)
Heat resistance, properties after:		4.6.9 250° ± 3°C (482° ± 5.4°F), 96 hours
Tensile strength, psi (MPa), minimum	2,000(13.8)	
Ultimate elongation, percent, minimum	200	
Dielectric strength, volts/mil (Kv/mm), minimum	500(12.7)	
Flammability	Non-flammable	2/
Corrosion	Non-corrosive	2/
Fluid resistance	No change	2/
Fungus resistance	No growth	2/

1/ Unless otherwise specified, the requirements, test procedures and conditions are for all classes.

2/ Because of their relative inertness, MIL-I-23053/11 sleevings are not normally subject to these requirements. However, the acquiring activity (see 6.2.1) may, at its option, specify performance of any or all of these requirements.

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Shelf life conditions: Supplier shall certify to storage at 40°C (104°F) for 4 years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: Heat shrinkable fluorinated ethylene propylene sleevings are intended for use in high temperature applications.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - MI
Navy - EC
Air Force - 99
DLA - GS

User activities:
Navy - SH

Preparing activity:
Navy - AS

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